JAN FEB MAR APR MAY JUN JULY AUG SEP OCT NOV DEC
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31

Problem 1 - FrogJmp

START
X = -1
Y = -1
D = -1
WHILE X is not between the range 1-1,000,000
    TRY
    INPUT X
    EXCEPT
    PRINT error message
WHILE Y is not between range 1-1,000,000 or Y is less than X
    TRY
    INPUT Y
    IF Y is less than X
    PRINT error message
    EXCEPT
    PRINT error message
WHILE D is not between the range 1-1,000,000
    TRY
    INPUT D
    EXCEPT
    PRINT error message
INVOKE function solution
jump-count = 0
    WHILE X is less than Y
        x increments by D
        jump-count increments by 1
RETURN jump-count
PRINT jump-count
END

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PROBLEM 2 - OddOccurencesInArray

START
INVOKE function solution
FOR the length of the array
paired_numbers = 0
FOR each element in the array
IF x is equal to y
paired_numbers = paired_numbers + 1
END IF
END FOR
IF paired_numbers is an odd value
unpaired_number = x
BREAK loop
END IF
END FOR
RETURN unpaired_number PRINT unpaired_number
END

PROBLEM 3 - PermMissingElem

```
START
INVOKE function solution
 N = length of array
    FOR each number from 1 to (N + 1)
        exists is False
            FOR the length of the array
                IF x is equal to y
                exists is True
                END IF
    END FOR
    IF exists is False
    missing_number = number
    RETURN missing_number
    END IF
END FOR
PRINT missing_number
END
```

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PROBLEM 4 - PermCheck

START
INVOKE function solution
CREATE empty set of elements
FOR the length of the array
CREATE range of numbers 1-N
END FOR
ADD range of numbers to empty set
FOR each number in set
IF the number is not in the array
RETURN 0
END FOR
RETURN 1
END
PRINT solution
END

PROBLEM 5 - FrogRiverOne

```
START
INVOKE function solution
CREATE empty set of elements
    FOR the length of the array
        ADD current leaf position to empty set
        IF length of empty set is equal to X
            RETURN k
        END IF
    END FOR
    RETURN -1
    PRINT solution
END
```